

Han Joon Byun

MACHINE LEARNING · OPTIMIZATION · FINANCIAL AI

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RESEARCH PROFILE

Ph.D. student at Seoul National University's Optimization and Financial Engineering Lab. Research focuses on tabular data, time series, evolutionary computation, and Transformers for effective, efficient, and interpretable financial prediction.

EDUCATION

Seoul National University Ph.D., Computer Science & Engineering	Expected Aug. 2027
Optimization and Financial Engineering Lab · Advisor: Prof. Byung-ro Moon	
Seoul National University M.S., Computer Science & Engineering	Aug. 2024
New York University B.S., Mathematics	Sept. 2018

RESEARCH & PROFESSIONAL EXPERIENCE

Agency for Defense Development Joint Research Project	2026
Contributed to interpretable-model research and transparent model behavior.	
PFCT Joint Research Project	2025
Developed a default-prediction model using tabular Transformers for financial data.	
Think Pool Joint Research Project	2022–2024
Developed AI-driven financial factors for quantitative finance.	
Republic of Korea Air Force Interpretation Officer	2019.06–2022.05

PUBLICATIONS & MANUSCRIPTS

Multi-Objective Evolution of Patch Tokens for Tabular Transformers
Han Joon Byun, Joon Huh, Jieun Yook, Byung-ro Moon GECCO 2026 · doi:10.1145/3795095.3805076
Evolutionary Feature Construction for Tabular Transformers via Gradient-Enhanced Attention Rollout
Han Joon Byun, Jieun Yook, Hanbin Koo, Byung-ro Moon GECCO 2026 Companion · doi:10.1145/3795101.3805282
CycleFormer: TSP Solver Based on Language Modeling
Jieun Yook, Junpyo Seo, Joon Huh, Han Joon Byun, Byung-ro Moon arXiv:2405.20042 · 2024 · Preprint
Evolutionary Pruning of Deep Convolutional Networks by a Memetic GA with Sped-Up Local Optimization and GLCM Energy Z-Score
Ha Hyun Cho, Han Joon Byun, Min Kee Kim, Joon Huh, Byung-Ro Moon GECCO 2023 Companion · doi:10.1145/3583133.3590604
RTSC: Return-Aware Token-Space Contrastive Transformer for Cross-Sectional Stock Return Prediction
Hanbin Koo*, Han Joon Byun*, Byung-ro Moon ICAIF 2026 · Under review · *Equal contribution
Label-Noise Decay Confounds Loss-Based Drift Attribution: A False-Positive Channel and a Noise-Aware Repair
Huh Joon, Han Joon Byun TMLR · 2026 · Under review

TEACHING EXPERIENCE

Head TA, Seoul National University Data Structures: Spring 2023, Fall 2024 & 2025
Algorithms: Fall 2023, Spring 2025 & 2026
Prepared course materials, supported students, coordinated TAs, and graded assignments and exams.